

Operator Worksheet

GPM / Tip Size	Nozzle Pressure	Hose	FL	Appliance	Elevation	PDP

Setups

Setup	GPM	Hose	FL	Nozzle	NP	Appliance	Elevation	PDP
First Floor Attack	185	1.75" × 200'	32	SB 15/16"	50	—	—	82
Third Floor Standpipe	185	3" × 50' → 2.5" × 100'	S 0.4 A 8.0	Fixed Fog	100	—	10	143

Operational Reference

Appliance Loss Guide		Additional Water Available	
APPLIANCE	ADD	INTAKE PRESSURE DROP	ADDITIONAL WATER AVAILABLE
Small Appliance (Wye, Reducer)	10 psi	0-10%	≈ 3× Current Flow
Large Appliance (Portable Hydrant, Water Thief)	25 psi	11-15%	≈ 2× Current Flow
Master Stream Device	25 psi	16-25%	≈ Current Flow
Standpipe System	25 psi	>25%	Less than Current Flow
Residential Sprinkler System	95 psi	Approximation only.	
Commercial Sprinkler System	150 psi		



Friction Loss Chart PSI per 100 ft

GPM	1.75"	2.5"	3"	5"
0	0.0	0.0	0.0	0.0
50	3.9	0.5	0.2	0.0
100	15.5	2.0	0.8	0.1
150	34.9	4.5	1.8	0.2
200	62.0	8.0	3.2	0.3
250	96.9	12.5	5.0	0.5
300	139.5	18.0	7.2	0.7
350	189.9	24.5	9.8	1.0
400	248.0	32.0	12.8	1.3
450	313.9	40.5	16.2	1.6
500	387.5	50.0	20.0	2.0
550	468.9	60.5	24.2	2.4
600	558.0	72.0	28.8	2.9
650	654.9	84.5	33.8	3.4
700	759.5	98.0	39.2	3.9
750	871.9	112.5	45.0	4.5
800	992.0	128.0	51.2	5.1
850	1119.9	144.5	57.8	5.8
900	1255.5	162.0	64.8	6.5
950	1398.9	180.5	72.2	7.2
1000	1550.0	200.0	80.0	8.0

1.75": C 15.5 2.5": C 2.0 3": C 0.80 5": C 0.08

Smoothbore References

Handline Smoothbore (50 PSI)		Masterstream Smoothbore (80 PSI)	
TIP	GPM	TIP	GPM
7/8"	161	1 1/4"	415
15/16"	185	1 1/2"	598
1 1/8"	266	2"	1063
1 1/4"	328		

Quick Reference

Common Formulas		Dry Standpipe Quick Reference	
FRICITION LOSS FL = C × (Q ÷ 100)² × (L ÷ 100)	SMOOTHBORE FLOW GPM = 29.7 × d² × √NP	THINGS TO REMEMBER - Riser supports 500 GPM - Outlet supports 250 GPM - PRVs may be present	THINGS TO ADD - Add 25 psi Standpipe Loss - Add 5 psi/floor Elevation Loss - Include Attack Friction Loss - Include Supply Friction Loss
SMOOTHBORE NOZZLE REACTION NR = 1.57 × d² × NP	FOG NOZZLE REACTION NR = 0.0505 × GPM × √NP		